

Applying NLP for Topic Modelling in a Real-Life Context

PureGym Customer Review Analysis
CAM DS C301 — Week 4 and 5 Topic Project

1. Overview of Approach

This project set out to analyse twelve months of customer review data collected from two platforms — Google Reviews and Trustpilot — for PureGym, one of the UK's largest budget gym operators. The central goal was to move beyond simple star ratings and uncover the specific themes, pain points, and emotional responses that drive negative customer experiences across the company's gym network.

The analysis followed a structured, multi-stage pipeline. The work began with data loading and cleaning, moved through exploratory frequency analysis and visualisation, then applied progressively more sophisticated techniques — BERTopic for semantic topic modelling, a BERT-based emotion classifier to detect anger, the Falcon-7B-Instruct large language model for topic extraction, and finally Gensim's LDA model as a validation step. At each stage, the output of one analysis informed the input and focus of the next, creating a coherent investigative pipeline rather than a set of isolated experiments.

Throughout the project, the analysis was always grounded in the business question: what is actually frustrating PureGym's members, and which specific locations are generating the most complaints? The technical choices — model selection, filtering criteria, prompt design — were all made in service of that practical question.

2. Description of Analysis

Data Loading and Preparation. Both datasets were loaded from CSV files and reduced to three columns each: the review text, the star rating, and the gym location. Column names were standardised across both sources to allow consistent processing. Missing review rows in the Google dataset were dropped entirely, while missing location values in Trustpilot were filled with 'Unknown' to avoid data loss on the review text itself.

Text Cleaning and Frequency Analysis. A custom cleaning function was applied to every review. It converted text to lowercase, stripped all punctuation and numeric characters using regular expressions, tokenised the sentence into individual words using NLTK's `word_tokenize`, and then removed standard English stopwords. This produced a clean list of meaningful tokens per review. Word frequency counts were computed using Python's Counter, and the top ten most common words were plotted as bar charts for both platforms. Word clouds were generated for a quick visual summary. Across both datasets, the most frequent terms in all reviews were 'gym', 'equipment', 'great', 'good', 'staff', and 'clean' — reflecting a mix of positive and operational language.

Negative Review Filtering. Reviews with a star rating below 3 were isolated as the negative subset. This produced a focused dataset of complaints from both Google and Trustpilot. The frequency and word cloud analyses were repeated on this subset. The shift was clear: positive words like 'great' and 'good' dropped out of the top results, replaced by words such as 'equipment', 'machines', 'membership', 'staff', 'get', 'time', and 'people'. The word 'equipment' appeared prominently in both negative datasets, suggesting that broken or inadequate equipment is a widespread and consistent frustration.

BERTopic Modelling — First Run. The negative reviews from locations common to both platforms were merged into a single list and passed to BERTopic. The model used the all-MiniLM-L6-v2 sentence transformer to embed each review into a semantic vector space, then applied UMAP for dimensionality reduction and HDBSCAN for clustering. The resulting topics were inspected via the interactive intertopic distance map, a bar chart of top words per topic, and a similarity heatmap. Topics discovered in this first run broadly mapped to equipment failures, facility cleanliness, staff behaviour, and membership administration.

Location Analysis — Identifying Problem Gyms. Negative review counts were computed separately for each dataset, then merged on gym location name and summed to produce a combined total. The top 30 locations by total negative reviews were identified and used as the basis for a second round of analysis. Word frequency and word clouds were re-run on this filtered set, followed by a second BERTopic run. Focusing on the most-complained-about gyms produced more concentrated topic clusters, confirming that the same themes — particularly equipment and overcrowding — are especially pronounced in the worst-performing branches.

Emotion Analysis. The pre-trained BERT model *bhadresh-savani/bert-base-uncased-emotion* was loaded via Hugging Face’s pipeline API and applied to every negative review in both datasets. The model classifies each review into one of six emotions: joy, sadness, anger, fear, love, or surprise. The emotion distribution across all negative reviews was visualised as a bar chart. Anger emerged as the dominant emotion, followed by sadness. All reviews classified as expressing anger were extracted and fed into a third BERTopic run, which highlighted the specific issues most likely to cause genuine frustration — in particular, gym overcrowding, equipment breakdowns, and problems with membership cancellation.

Falcon-7B-Instruct Topic Extraction. A sample of five negative Google reviews was passed to the Falcon-7B-Instruct large language model with the following prompt prepended to each review: *‘In the following customer review, pick out the three main topics. Return them as a numbered list.’* The model successfully identified themes such as overcrowding, poor equipment availability, inadequate weight areas, management quality, and customer service failures. These extracted topic strings were then collected and fed into a fourth BERTopic run using a customised UMAP and HDBSCAN configuration with lower minimum cluster sizes, producing refined topic clusters from the AI-generated summaries.

Gensim LDA — Validation. As a final step, Latent Dirichlet Allocation was applied to all negative review tokens from both platforms combined. A dictionary was built from the tokenised reviews, a bag-of-words corpus was constructed, and an LDA model was trained with 10 topics over 10 passes. The pyLDAvis visualisation was used to inspect topic separation. The ten LDA topics confirmed themes already identified by BERTopic, including facility cleanliness, equipment maintenance, access and administration issues, and general overcrowding and staff problems.

Table 1: LDA Topic Summary (Selected Topics)

Topic	Top Words	Inferred Theme
0	gym, changing, showers, water, shower, dirty, staff	Facility hygiene
1	gym, equipment, machines, weights, broken, use	Equipment condition
2	gym, get, day, email, pin, help	Access and administration
3	gym, membership, cancel, month, pay, money	Membership and billing
4	gym, busy, people, time, peak, use	Overcrowding

3. Insights Identified

Across all analytical techniques applied in this project, four major themes emerged consistently as the primary drivers of negative customer experience at PureGym. The convergence of these themes across different methods — word frequency, BERTopic, emotion analysis, Falcon extraction, and LDA — gives strong confidence that they represent genuine, systemic issues rather than isolated complaints.

Equipment Availability and Maintenance. The word ‘equipment’ was the second most frequent term in both the Google and Trustpilot negative review datasets. BERTopic consistently produced a cluster centred on equipment-related language — broken machines, insufficient weights, machines out of order, and long waits to access certain equipment. Falcon’s topic extraction from individual reviews also flagged ‘poor equipment availability’ as a recurring concern. For a gym chain that positions itself as high-quality and affordable, broken or scarce equipment represents a direct failure of the core value proposition. Members who cannot complete their intended workouts due to equipment issues are unlikely to renew.

Overcrowding and Gym Capacity. Multiple BERTopic runs surfaced topics relating to overcrowding — with words like ‘busy’, ‘crowded’, ‘peak’, ‘wait’, and ‘full’ appearing in prominent clusters. Falcon’s review of one specific review explicitly identified ‘overcrowding’ as the primary topic. Several negative reviews expressed frustration that the gym had become unusable at peak hours, and some specifically linked the overcrowding to PureGym’s aggressive membership growth strategy. This is a structural challenge: the low-cost model depends on high membership volume, which naturally creates capacity pressure at popular locations.

Staff Behaviour and Customer Service. Staff-related language appeared in both the all-reviews frequency analysis and in negative-specific clusters. Words such as ‘staff’, ‘rude’, ‘unhelpful’, ‘management’, and ‘attitude’ featured across BERTopic topics and LDA’s output. The emotion model’s identification of anger as the dominant negative emotion is likely connected to staff interactions — poor service tends to generate a stronger emotional reaction than passive dissatisfaction with facilities. Falcon’s extraction from one review explicitly flagged ‘poor management and staff’ as a key topic.

Membership Administration and Billing. The Trustpilot negative dataset showed a particularly strong signal around the word ‘membership’, which ranked third in the top negative words list with 724 occurrences. LDA Topic 3 was clearly centred on billing and cancellation issues — ‘cancel’, ‘month’, ‘pay’, ‘money’. Trustpilot, as a platform often used by consumers to report service failures, tends to attract reviews from members who have had administrative disputes, which explains why this theme is stronger there than on Google. Nevertheless, membership cancellation difficulties and unexpected charges are a well-documented pain point in the budget gym sector and clearly affect PureGym.

Location-Specific Patterns. The analysis of the top 30 worst-performing locations revealed that the problems identified above are not evenly distributed. Certain gym branches attract a disproportionately high number of complaints across both platforms. When the same location appears in both the Google and Trustpilot top-20 complaint lists, this is a strong signal of a localised management or facility issue that warrants direct investigation. The topic analysis on the top 30 locations produced tighter and more consistent BERTopic clusters than the broader first run, confirming that the worst branches share specific recurring problems rather than random dissatisfaction.

4. Conclusions and Recommendations

The combined application of BERTopic, emotion classification, LLM-based topic extraction, and classical LDA produced a consistent and credible picture of what is driving negative reviews at PureGym. The fact that all four methods converged on the same four themes —

equipment, overcrowding, staff, and billing — without any prior labelling or supervision demonstrates the power of NLP for large-scale feedback analysis.

For PureGym’s leadership team, the most actionable insights from this analysis are: first, implement a proactive equipment maintenance and reporting system so that broken machines are identified and repaired faster; second, introduce capacity management measures at the most overcrowded branches — for example, booking systems for peak hours; third, invest in customer-facing staff training with a particular focus on conflict resolution and member retention; and fourth, simplify the membership cancellation process to reduce billing disputes, which are a significant source of anger especially among Trustpilot reviewers.

More broadly, this project demonstrates that a well-structured NLP pipeline — combining semantic embeddings, emotion detection, and LLM-assisted extraction — can extract richer and more actionable business intelligence from customer reviews than simple keyword counting alone. The multi-stage approach, where each layer of analysis builds on and refines the previous one, is particularly effective for large, noisy text datasets of the kind that businesses routinely collect but rarely analyse deeply.

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